FUNDS CONTROL

USER MANUAL

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About this document

This document describes the maintenance features of the Funds Control (FCL) application and explains how to use the application menus and submenus.

Purpose

This manual helps you understand the capabilities of the FCL application, as well as guides you through the usage procedures.

Scope

This document covers the end-to-end usage of the Funds Control application.

Structural makeup of this document

The document is organized as follows:

About this – Describes the purpose, scope, and document structure.

System overview and capabilities – Provides an overview of the FCL application, including its features and graphical user interface (GUI).

FCL workflow - Explains the FCL process flow for providing Pay/No Pay/Refer decisions for funds check requests.

General characteristics - Describes how FCL generally handles workflow anomalies, like error and system messages.

System access procedure - Explains how to sign in, use the menus, and sign out.

Global maintenance - Covers audit trails and the maker-checker concept.

Detailed description – Workflow module - Provides an overview of the Workflow module, including its scope, job flow, and global maintenance.

Error! Reference source not found. - Provides detailed descriptions of the Workflow module menus, procedures, and fields. This chapter is the heart of the manual.

Error! Reference source not found. - Provides a detailed description of the BPS module menus, procedures, and fields.

Error! Reference source not found. – Lists abbreviations and acronyms used in this manual.

System overview and capabilities

This chapter provides an overview of the FCL application, its features, and its graphical user interface (GUI). Read this chapter to familiarize yourself with the application before navigating its screens.

Browser settings

Before you launch the application, configure the following settings in Internet Explorer (IE).

#	IE Version	Navigation Path	IE Settings
1	IE 6	Tools > Internet Options > Security tab > Internet icon > Custom Level	In the **Scripting** section, under **Allow paste operations via script**, select **Disable**.
2	IE 7		In the **Scripting** section, under **Allow Programmatic clipboard access**, select **Disable**.
3	IE 8		In the **Scripting** section, under **Allow Programmatic clipboard access**, select **Disable**.

System overview

The Funds Control Application (FCL) provides a Pay/No Pay/Refer decision for funds check requests based on the availability of adequate funds in a debit account. FCL arrives at a decision by analyzing the real-time balances and limits in a debit account, as well as all associated accounts when the debit account is part of a complex account structure.

The FCL system is invoked by the Global Pay Plus (GPP) system when GPP receives a payment request. GPP sends a request to FCL to determine whether adequate funds are available in the debit account. Based on fund availability, FCL returns a Pay or No Pay decision. When manual intervention is required, FCL returns a Refer decision.

Features of FCL

FCL has the following key features:

- User authentication for secure system access
- Multi-office access to the FCL application
- Pay/No Pay decisions for payment requests

- Handling of both simple and complex account types
- Calculation engine for balance computation in complex accounts
- Referral process for cases that require manual intervention
- Automatic retry of transactions in the Refer queue to reduce manual handling
- Ability to resubmit a transaction with changes to core data fields
- Audit trails for all FCL system events
- Ability to cancel a request already initiated in FCL
- Earmark and exposure functionality
- Storage of feed files received from the Backend Exchange Module (BEM)
- GUI for maintaining static data and business rules

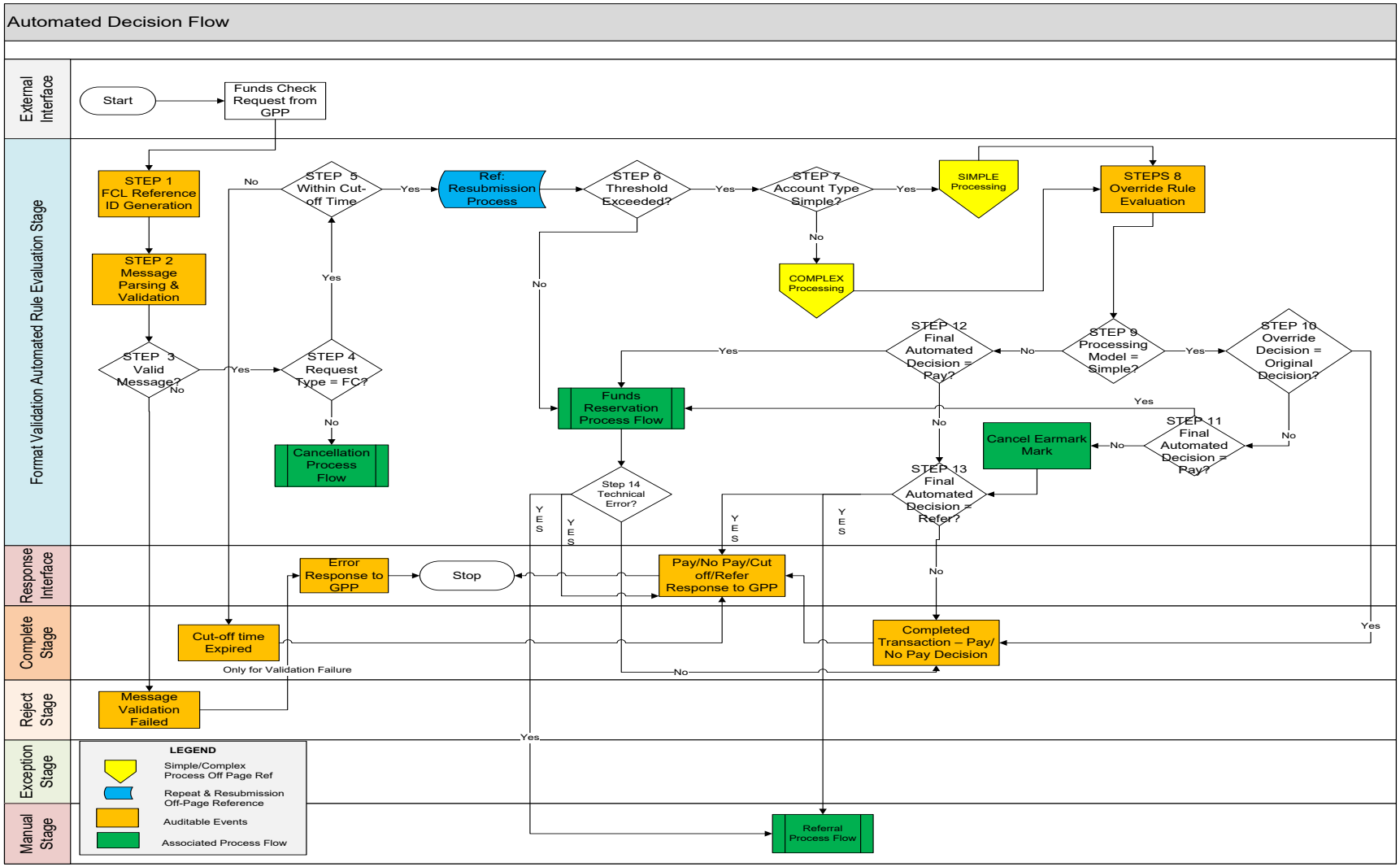
FCL workflow

This chapter describes the FCL workflow.

Product workflow - Detailed description

The diagram below gives a visual representation of the product workflow.

Main FCL Process



The following steps describe the main FCL process:

1. FCL generates a unique Funds Control Reference ID for each incoming message from GPP and introduces the record into the first stage of the Funds Control process.

2. FCL parses the message and validates its format and content. A typical incoming message includes the following information:

    - Source system unique ID
    - Source system reference
    - Accounting system ID
    - Account ID
    - Debit currency
    - Amount
    - Value date
    - MOP
    - Channel
    - Cut-off time

3. FCL checks whether the information in the incoming request is valid for each FCL reference ID. If the message is invalid, FCL moves the request to the Reject stage with a status of Validation Failed.

4. FCL reads the request to determine the type. Incoming requests from GPP can be one of the following types:
   Funds Check Request (fresh or repeat request, or resubmission with changed core fields) – This type moves to step 5.
   Cancellation of Funds Check Request – This type follows the cancellation processing steps.

5. FCL validates the cut-off time for each Funds Check request. Requests within the cut-off time proceed through Resubmission and Repeat Submission validation before moving to step 6. Requests past the cut-off time are moved to the Completed stage with a status of Cut-off Time Expired.

6. FCL performs a threshold validation for each qualified Funds Check request. If the request amount falls within the threshold limit, FCL returns an Automatic Pay decision and routes the request to the Funds Reservation process. If the amount does not qualify, FCL proceeds to step 7.

7. FCL checks whether the debit account is a simple or a complex account. Simple accounts follow the Simple Account processing model (see section 3.1.2). Complex accounts follow the Complex Account processing model (see section 3.1.3).

8. After simple or complex processing, FCL determines a Pay/No Pay decision and evaluates the request against the override rules in the Rule Engine.

Simple processing override validation

For simple accounts, FCL checks the override parameters and validates the decision against the funds check request. Example: If Decision = No Pay and SBU = Wealth, then Override Decision = Refer. The process returns a final automated decision, which may differ from the original decision.

Complex processing override validation

For complex accounts, FCL evaluates override parameters to arrive at the final Pay/No Pay/Refer decision. Example: If Previous Decision = No Pay, SBU = Personal, Customer ID = 1234, then Final Automated Decision = Pay.

9. After the Override Rule evaluation, FCL checks the processing model. If Simple, FCL proceeds to step 10. If Complex, FCL proceeds to step 12.

10. FCL compares the Override decision to the Original decision. If they match, FCL moves the transaction to the Completed stage. If they differ, FCL proceeds to step 16 for further evaluation.

11. If the final automated decision is Pay, FCL moves the request to the Funds Reservation process. If the original decision was No Pay (and is now overridden to Pay), FCL submits a forced earmarking request to the backend system through BEM. After BEM responds, the request moves to the Completed stage. If the override decision is not Pay, FCL moves the request to the Cancel Earmark process. After earmark cancellation, the request proceeds to step 18.

12. For complex accounts with a final automated decision of Pay, FCL moves the request to the Funds Reservation process for earmarking or exposure marking. After that process completes, requests move to the Completed stage. Requests with a decision other than Pay proceed to step 13.

13. FCL checks whether the final automated decision is Refer. If yes, FCL moves the request to the Manual stage for manual processing and invokes the Referral Setup process. If no, FCL moves the request to the Completed stage with a No Pay decision. If a request with the same Source System Unique ID already exists in the Wait stage, the request is treated as a resubmission.

14. If a technical error occurs during the Funds Reservation process, FCL routes the request to the Referral stage with a status of Error.

Simple account processing model

15. Processing model flagging: FCL checks whether the debit account can be processed as a simple account by verifying whether the accounting system ID supports the simple account processing model. Simple account processing is configured at the office level by specifying

the accounting system IDs that support it. If the accounting system ID does not support simple processing, FCL flags the request as Processing Model = Complex.

16. **Bypass Accounting System Call**: FCL checks whether the Bypass Accounting System Call flag is set to Y for the office and accounting system. If Y, FCL proceeds to step 3. If N, FCL proceeds to step 6.

17. **Evaluate Contingency Rule**: FCL checks whether the Evaluate Contingency Rules flag is on. If Y, FCL proceeds to step 4. If N, FCL returns a Refer decision and routes the request to the Referral process.

18. **Contingency rule evaluation - whitelist lookup**: FCL checks whether the debit account is in the whitelist. If found, FCL returns an Auto Pay decision. If not found, FCL returns a Refer decision and routes the request to the Referral process.

19. **Auto Pay decision**: FCL routes the request to the Funds Reservation process for earmarking or exposure marking. This step rejoins the Automated Decision main flow at Funds Reservation.

20. **Processing model flag = Simple?** If the Bypass Accounting System Call flag is N, FCL checks whether the processing model is Simple. If Simple, FCL proceeds to step 7. If Complex, FCL proceeds to step 9.

21. **Authorize and Earmark Request Generation**: FCL generates a new Earmark Reference Number and submits an Authorize and Earmark request to BEM. BEM forwards the request to the backend accounting system. FCL waits for a response for a configurable period. If no response is received, FCL returns a Technical Error decision and proceeds to step 8.

22. **Technical Error?** BEM returns the decision to FCL within the configured wait period. If a technical error occurs while receiving the response, FCL marks the request as Error and routes it to step 3. For non-error transactions, the flow rejoins the Automated Decision main flow from step 8 onwards.

23. **Get Multi Account Balance Request**: If the processing model is Complex (that is, the accounting system does not support Authorize and Earmark), FCL generates a Get Multi Account Balance request through BEM. The Calculation Engine triggers this request.

INFO: SBU (Strategic Business Unit) information retrieval: For simple transactions with a No Pay decision, BEM makes an additional call to CDI to obtain the SBU value for the debit account and includes it in the response to FCL. For simple transactions with a Pay decision, BEM does not return the SBU value because there are currently no override rules that would change a Pay decision to Refer.

24. **Technical Error?** BEM returns the requested balances to FCL. If a technical error occurs or a timeout is encountered, FCL marks the request as Error and routes it to the Stored Balance Retrieval process. For non-error requests, FCL proceeds to step 12.

25. **Stored Balance within age limit?** FCL evaluates the age of the retrieved balances. If the stored balances for error transactions are within the age limit, FCL proceeds to step 12. Otherwise, FCL routes the transaction to step 3.

26. **Receive and store balance against transaction**: FCL stores the received balances and limits against the Funds Check request. Because simple accounts do not have a limit structure in FCL's master data, the Limit Master is not updated.

27. **Calculate simple availability**: FCL feeds the stored balances and limits into the Calculation Engine, which applies the Simple Availability Calculation formula.

28. **Business rule evaluation**: FCL invokes the Business Rule Evaluation process. The calculated availability is fed into the Business Rule Engine, which returns a Pay/No Pay decision. The flow then rejoins the main Automated Decision flow from step 8.

Complex account processing model

29. Processing model flagging: FCL flags requests with a complex debit account as Processing Model = Complex.

30. **Bypass Accounting System Call**: FCL checks whether the Bypass Accounting System Call flag is set to Y for the office and accounting system. If Y, FCL proceeds to CN step 4. If N, FCL proceeds to CN step 3. In a complex scenario where account structure entities span multiple accounting systems: FCL identifies the accounting system for the debit account and checks the flag. If Y, FCL skips the interface call. If N, FCL calls the backend accounting system through BEM, which retrieves balances for all entities in the structure. If any accounting system is unavailable, BEM returns a Technical Error to FCL, and FCL follows the Complex process from CN step 7 onwards.

31. **Feed failure check**: FCL receives details of complex account structures and associated limits in a daily end-of-day feed from UKBA, SOLD, and MMS. If a feed is not received within the defined service level agreement (SLA) or cannot be uploaded, FCL sets a Feed Failure flag at the Office + Accounting System level. FCL checks this flag for all Funds Check requests with Account Type = Complex. If the flag is set, FCL proceeds to step 4. If not, FCL checks whether the debit account belongs to an MOF structure. If not MOF, FCL proceeds to step 7. If MOF, FCL checks the MMS feed flag. If the MMS flag is set, FCL proceeds to step 4. If not, FCL proceeds to step 7.

32. **Evaluate Contingency Rule**: FCL checks whether the Evaluate Contingency Rules flag is on. If Y, FCL proceeds to step 5. If N, FCL returns a Refer decision and routes the request to the Referral process.

33. **Contingency rule evaluation - Whitelist lookup**: FCL checks whether the debit account is in the whitelist. If found, FCL returns an Auto Pay decision. If not found, FCL returns a Refer decision.

34. **Auto Pay decision**: FCL routes the request to the Funds Reservation process for earmarking or exposure marking. This step rejoins the Automated Decision main flow at Funds Reservation.

35. **Get Multi Account Balance Request**: FCL generates a Get Multi Account Balance request through BEM for complex account structures. The Calculation Engine triggers this request. For details, see IDD F06 and the Calculations FDD.

INFO: SBU information retrieval for complex transactions: (a) New accounts registered via a complex structures feed do not have an SBU value initially. The SBU is populated the next business day through the overnight Accounts Update feed. (b) New accounts manually registered in FCL receive the SBU value through the account validation call to BEM, if the call succeeds. Otherwise, the SBU is populated the next business day. (c) During the first calculation iteration, FCL checks whether the Facility record for the debit account already has an SBU value. If yes, FCL copies it to the transaction and sets the SBU required flag to N in the BEM MAE request. If no, FCL sets the flag to Y. If the MAE call succeeds, FCL stores the returned SBU on the transaction and the Customer Data record in the Facility Master. (d) For retried complex transactions, the SBU is already populated, so FCL sets the SBU required flag to N.

36. **Technical Error?** BEM returns the requested balances to FCL. If a technical error, an incomplete balance response, or a timeout occurs, FCL marks the request as Error and routes it to the Stored Balance Retrieval process. For non-error requests, FCL proceeds to step 9.

37. **Stored Balance within age limit?** FCL evaluates the age of the retrieved balances. If the stored balances are within the age limit, FCL proceeds to step 9 of the Complex process flow. Otherwise, FCL routes the transaction to step 3.

38. **Receive and store balance:** FCL stores the balances returned by BEM or retrieved from storage, and feeds them into the Calculation Engine.

39. **Calculation for balance availability:** FCL feeds the received balances into the Calculation Engine to evaluate the arithmetic formulas and determine net balance availability at each entity level (account, group, CAS, MASS, or MOF). The Calculation Engine returns available balances at each entity level to the Rule Engine. For details, see the Calculations FDD.

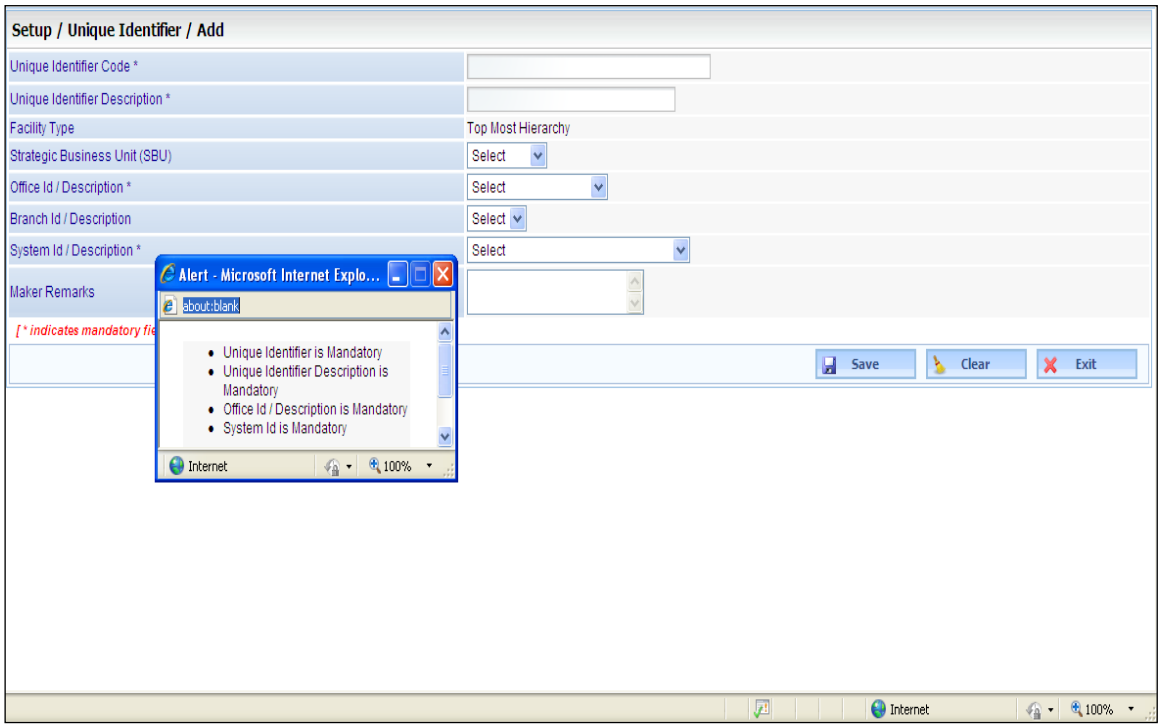
40. **Business rule evaluation**: FCL invokes the Business Rule Evaluation process. The system evaluates configured business rules and returns a Pay/No Pay/Refer decision. The Calculation Engine passes all computed formula codes for each entity in the structure to the Business Rules Engine. After the decision is reached, the flow rejoins the main FCL process from step 8.

General characteristics

GUI error handling

If you enter incorrect or invalid input while using the FCL application, the system displays an error message or warning with a description of the issue.

Example: If you try to submit a transaction without entering values in the mandatory fields, the system displays a message that lists the unfilled mandatory fields.



Help facility

Not applicable for current release.

System access procedure

This chapter describes how to sign in to the FCL application, navigate the menus, and sign out.

Access procedure

Follow these steps to start and sign in to the FCL system.

To access the FCL application:

1. Sign in to the system using the Tivoli Access Manager (TAM)/WebSEAL application.

2. Click the Funds Control Application link.
   You are redirected to the Armor application. Armor authenticates you based on the roles received from TAM/WebSEAL.
   If authentication succeeds, the FCL application selection page appears and lists the applications your group can access.

3. Select the link for the application you want to access.
   Example: Select **Workflow**. The Workflow application main screen appears.
   If you have access to multiple applications and want to switch, select the **Logout** link in the upper-right corner of the current application, then select the link for the application you want.

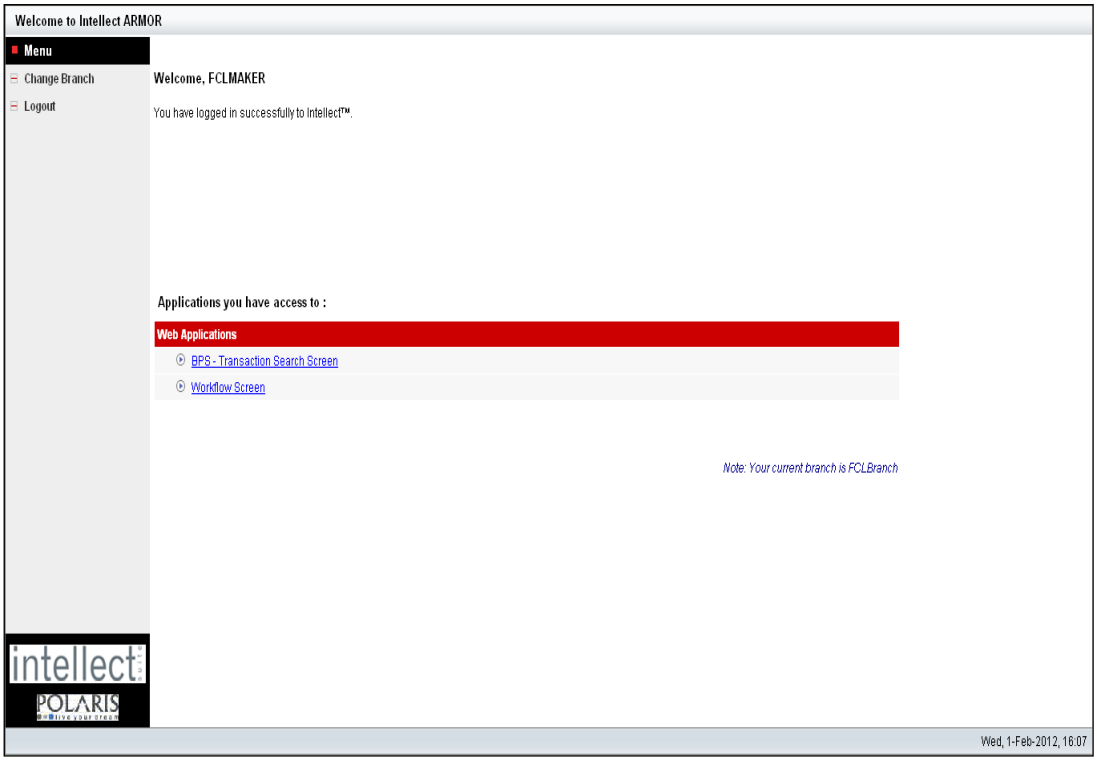
If authentication fails, the system displays a technical error message.

Change FCL branch

In FCL, applications are organized by branch. The Armor application preconfigures two branches: ARMADMBRANCH (which includes the ArmorWebAdmin application) and FCL (which includes the FCLadmin and FCL applications). These branches are mapped to groups. If your group is mapped to both branches, you can switch between them.

To change the branch:

1. Sign in to the FCL application.
   The Armor welcome page appears.



2. In the left pane, select **Change Branch**.
   The Change Branch page appears.

3. Select the required branch from the drop-down list.

4. Select the **Change** link.
   The Welcome page appears and lists the applications mapped to the selected branch.

Sign in using FCL dummy page

This procedure applies to the current FCL setup in which TAM/WebSEAL is not integrated with FCL. A temporary login page is provided.

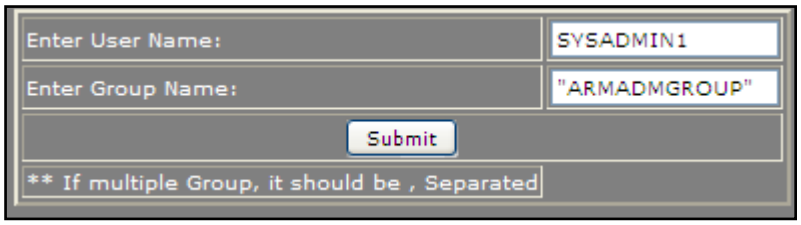
To sign in to the FCL application using the dummy page

1. Go to the following URL: ***http://<HostName>.polaris.co.in:<Bootstrapport>/AAL/***
   For instance, http://pslaix05.polaris.co.in:10088/AAL/.
   The FCL login page appears. By default, the user is SYSADMIN1 and the group is ARMADMGROUP.

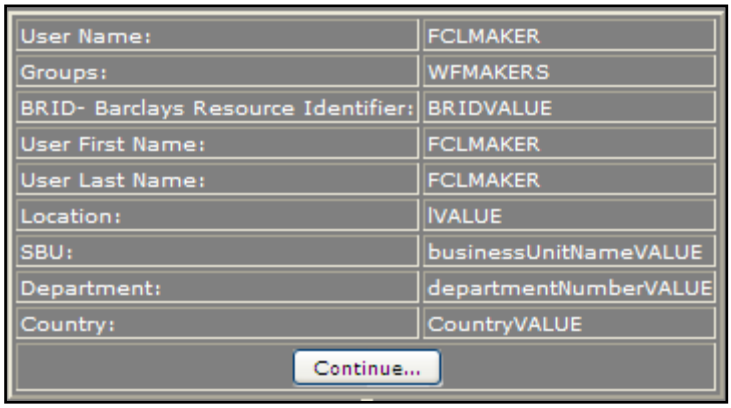
2. In the **Enter User Name** box, type your name.

3. In the **Enter Group Name** box, select the name of the Group you are mapped to.
   The group must have the appropriate rights to access the Workflow application.

4. Click **Submit**.



5. In the FCL login details page that appears, click **Continue**.



6. In the FCL application selection page that appears, click the link for the application you want to access.
   Example: Select **Workflow**. The Workflow application main screen appears.